ModelIt!

Student Activity Pack

Why Do Some Countries Run Out of Water?

Modeling Water Stress from Rainfall, Population, and Consumption

Grade 06L2 | NGSS MS-ESS3-1, MS-ESS3-4 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where does the water that comes out of your tap originally come from?

2 Why do you think some places in the world do not have enough clean water?

3 If Earth is 70% covered in water, why would any country run out of water?

4 Draw what you think happens to rainwater after it hits the ground.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Rainfall Pattern | Population Density | Aquifer Recharge | Water Demand | Water Stress Index

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Sustainable Balance

Moderate rainfall, low population density

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Growing Demand

Constant rainfall, population increasing

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Drought Crisis

Very low rainfall, moderate population

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Climate Change Effect — Long-term shifts in rainfall patterns and temperature that can permanently reduce water supply in some regions and increase it in others

Agricultural Water Use — Water consumed by farming, which typically accounts for 70% of all freshwater withdrawal globally and is often the largest demand on aquifers

Water Recycling Rate — The percentage of used water that is treated and returned to the supply system, which can significantly reduce net demand on freshwater sources

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If Earth has the same amount of water it has always had, why are some countries running out?
Use evidence from your simulation to support your answers.

1. **Why is water stress a local problem even though Earth has a global water cycle?**

2. **How is the time mismatch between aquifer recharge and human extraction the root of water crises?**

3. **What would happen to your model if a country doubled its population but also cut per-capita consumption in half?**

4. **Why might two countries with the same rainfall have very different Water Stress Indexes?**

5. **How does your model show that water stress is about balance, not just about supply?**

8 STEM Challenge: Design a Water Management Plan

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YOUR ENGINEERING MISSION

Design a sustainable water management plan for a fictional growing city that prevents water stress from exceeding critical levels over the next 50 years.

THE SCENARIO

A rapidly growing city in a semi-arid region currently has moderate water stress. Population projections show the city will triple in size over the next 50 years while climate models predict a 15% decrease in annual rainfall. Your environmental engineering team must design a comprehensive water management strategy that keeps the Water Stress Index below crisis level.

Engineering Design Thinking

1. WHAT CONSERVATION MEASURES WILL REDUCE PER-CAPITA WATER CONSUMPTION?

2. WHAT INFRASTRUCTURE (RAINWATER HARVESTING, WASTEWATER RECYCLING, DESALINATION) WILL SUPPLEMENT NATURAL SUPPLY?

3. HOW WILL YOU PRIORITIZE WATER ALLOCATION BETWEEN AGRICULTURE, INDUSTRY, AND HOUSEHOLDS?

4. HOW WILL YOU MONITOR THE WATER STRESS INDEX OVER TIME AND TRIGGER EMERGENCY MEASURES IF NEEDED?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Hydrologists and Water Resource Engineers study water systems and design sustainable management plans to ensure communities have clean water. They earn $60,000-$115,000/year.